



SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY

COMPUTER SCIENCE AND IT DEPARTMENT

COURSE INFORMATION SHEET

(For Lab Based Course)

Session:	Fall-2024
Course Title:	Web Engineering
Course Code:	CS-311L
Credit Hours:	1
Semester:	5th
Pre-Requisites:	Programming Fundamentals
Instructor Name:	Sir Rahil Usman
Email and Contact Information:	rausman@ssuet.edu.pk , 03313331242
WhatsApp Group	Yes
Office Hours:	09:00 AM To 5:00 PM
Mode of Teaching:	Synchronous/Asynchronous/ Hybrid/Blended

COURSE OBJECTIVE:

The goal of the course is to give knowledge about web engineering concepts, principles process, methods and models. How to model design for web applications and what models' perspective are useful for construction. Familiarize with Web application development software tools and environments currently available on the market. Build Web Applications that are scalable flexible to modify and easy to manage.

COURSE OUTLINE:

Web programming languages, Design principles of Web based applications, Web platform constraints, Web standards, Responsive Web Design, Web Applications, Browser/Server Communication, Storage Tier, Cookies and Sessions, Input Validation, Full stack state management, Web App Security, Large scale applications, Performance of Web Applications, Enabling Technologies for adoption of cloud services and future Data Centers, Web Testing and Web Maintenance.

COURSE OUTCOMES:

CO No.	Course Outcomes (COs)
1	Understand the processes, concepts and standards related to the web engineering.
2	Produce static and responsive designs for web application.
3	Develop dynamic web application.
4	Apply techniques to resolve security threats in web applications.
5	Apply web standards for web application development
6	Analyze and solve small-scale web engineering problems
7	Use modern engineering tools necessary for web engineering practice



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GRADING POLICY:

Assessment Tools	Percentage
Lab Manual	30%
Viva/Demonstration	30%
Lab Exam	40%
TOTAL	100%

Recommended Book:

- **Web Engineering the Discipline of Systematic Development of Web Applications** by Gerti Kappel, Birgit Proll, Siegfried Reich, Werner R. ISBN 13: 9780470015544

Reference Books:

- **Web Engineering: Modelling and Implementing Web Applications (Human–Computer Interaction Series)** 2014 Edition by Rossi ISBN 978-1-84628-923-1
- **Current Trends in Web Engineering** ICWE 2018 International Workshops, 2018



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LAB PLAN

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Week No.	Lab Date	Objective	Required Reading
1	14-10-2024 to 18-10-2024	HTML, CSS	Lab Manual Pages 1 - 8
2	21-10-2024 to 25-10-2024	Basic JavaScript	Lab Manual Pages 9 – 14
3	28-10-2024 to 01-11-2024	Document Object Model [DOM]	Lab Manual Pages 15 - 18
4	04-11-2024 to 08-11-2024	Browser Object Model [BOM]	Lab Manual Pages 19 - 22
5	11-11-2024 to 15-11-2024	Intermediate JavaScript	Lab Manual Pages 23 – 27
6	18-11-2024 to 22-11-2024	Bootstrap/SASS	Lab Manual Pages 28 – 32
7	25-11-2024 to 29-11-2024	ES5\ES6\ESNEXT\Typescript	Lab Manual Pages 33 – 37
8	Mid Term Examination (02-12-2024 to 07-12-2024)		
9	09-12-2024 to 13-12-2024	JavaScript- The Good Parts	Lab Manual Pages 38 – 40
10	16-12-2024 to 20-12-2024	Bundling / Webpack	Lab Manual Pages 41 – 43
11	23-12-2024 to 27-12-2024	React Class / Functional	Lab Manual Pages 44 – 47
12	30-12-2024 to 03-01-2025	React / Redux /Context	Lab Manual Pages 48 – 52
13	06-01-2025 to 10-01-2025	React Router	Lab Manual Pages 53 – 59



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14	13-01-2025 To 17-01-2025	Formik Forms + Material UI	Lab Manual Pages 60 – 62
15	20-01-2025 to 24-01-2025	E2E CRUD App Development using MongoDB, NodeJs	Lab Manual Pages 63 – 70
16	Final Examination (28-01-2025 to 08-02-2025)		